

Dobson Dunavant

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Skills

Languages: Python, Bash, TypeScript, JavaScript, SQL, JSON, YAML, HTML, CSS

Cloud/DevOps Tools: AWS, Azure, Terraform, AWS CDK, Docker, Kubernetes, Git, GitHub Actions, GitOps, Datadog

AWS Services: VPC, EC2, S3, RDS, Lambda, CloudWatch, IAM, Security Groups, Network Firewall, WAF, Shield, Site-to-Site VPN, ECS Fargate, CodePipeline, MSK, EKS

Frameworks/Libraries: React, Node.js, Express, Next.js, GraphQL, Axios, Jest, React Testing Library, K6

Databases: PostgreSQL, RDS Aurora, MySQL, Redis, MongoDB, Redshift

Experience

Cloud Systems Engineer, Savvas Learning Company – Remote, USA

Dec 2023 – Present

- Architected 0-downtime GCP-to-AWS cloud migration using Terraform and AWS CDK Infrastructure as Code, transitioning App Engine to ECS Fargate, Cloud Storage to S3, and BigQuery to Redshift
- Built FERPA-compliant cloud networking using AWS Site-to-Site VPN and Terraform, enabling secure cross-cloud transfers of sensitive student data between multi-cloud environments
- Deployed Terraform and Ansible-based Infrastructure as Code managing multi-environment AWS infrastructure including VPC, Security Groups, Network ACLs, and route tables across production systems
- Engineered AWS WAF solution with multi-layered rules and whitelisting logic to block malicious traffic while ensuring legitimate signed traffic passes through, protecting production APIs
- Established Datadog observability platform using AWS CDK and Terraform to deploy monitoring agents, custom alerts, and dashboards, improving detection by 40% and reducing costs \$53,000+ annually
- Orchestrated 0-downtime blue-green deployments across ECS Fargate services using AWS CodePipeline and CodeDeploy with GitHub Actions CI/CD integration, linear traffic switching, and automated rollbacks
- Led 3 major cloud infrastructure epics from planning to completion, including 0-downtime RDS PostgreSQL 12 to Aurora PostgreSQL 16 migration with minimal senior oversight
- Strengthened application security by resolving critical DAST vulnerabilities, implementing Auth0-based system lockdown protocols, and configuring enhanced HSTS security headers
- Architected multi-region AWS ElastiCache caching layer with automated failover using Terraform, reducing API latency by 29% during peak traffic periods and improving system resilience
- Built end-to-end data pipeline using AWS Kinesis Firehose streaming CloudWatch application logs to Redshift, creating materialized views aggregating disparate log data for analytics
- Decoupled cloud-native microservices using AWS SNS and SQS for asynchronous messaging, orchestrating multi-step workflows with AWS Step Functions to reduce inter-service coupling and improve system reliability
- Developed serverless microservice for real-time student course-progress tracking using AWS Lambda and DynamoDB, reducing p99 query latency by 23% and supporting 450+ peak concurrent sessions
- Architected modular k6 load testing suite with Datadog StatsD integration and Docker-based orchestration, creating reusable test patterns that democratized performance testing across engineering teams
- Identified and eliminated critical performance bottleneck in student dashboard API using k6 load testing, replacing expensive LEFT JOIN with inverse relationship lookups to reduce p95 response time from 5.8s to 795ms
- Led planning and development of semester-long course epic requiring multiple new full-stack features, completing deployment in 6 weeks, 2 weeks ahead of deadline, by coordinating parallel workstreams
- Developed full-stack solutions for Outlier.org educational platform using React, Node.js, Python, AWS, and GitHub Actions CI/CD in Agile microservices environment
- Built SQL-procedure-based PII scrambling solution allowing developers to pull anonymized production data subsets locally, reducing bug replication time from hours to minutes
- Redesigned database architecture by unifying disparate data sources, establishing schema relationships in PostgreSQL and RDS Aurora, implementing migration patterns for over 7TB of course data

- Authored 10+ comprehensive technical documentation pieces in Confluence covering AWS architecture, cloud migration strategies, and Infrastructure as Code patterns, improving developer onboarding
- Created and delivered 4+ technical knowledge-sharing sessions on AWS architecture, cloud security best practices, and cloud/database migration strategies for engineering teams

Software Engineering Co-Op, IFA Group – Ladson, SC

Aug 2023 – May 2024

- Led team of 4 developing containerized warehouse inventory system using Docker, React, Flask, AWS, and MySQL, delivering real-time inventory tracking through RESTful API integration with SAP endpoints
- Authored System Requirements and Design Specification documents collaborating with warehouse stakeholders, establishing clear technical architecture and deployment strategy for cloud-based solution
- Delivered end-to-end cloud-deployed system within 6 months using Terraform Infrastructure as Code and GitHub CI/CD, enabling real-time inventory awareness across multiple warehouse locations

Software Engineering Intern, Savvas Learning Company – Remote, USA

May 2023 – Aug 2023

- Developed full-stack solutions for Savvas Realize K-12 platform using Angular micro-frontends, GraphQL, RxJS, Spring Boot, MySQL, implementing software engineering best practices for testing and deployment
- Optimized GraphQL API queries by refactoring data-fetching logic and eliminating over-fetching, reducing payload sizes and improving application performance across distributed microservices architecture
- Won summer hackathon by leading development of PowerBI unique student insights dashboard, demonstrating data analysis and visualization skills for educational technology stakeholders
- Improved API test coverage by 19% through automated Karate testing framework CI/CD integration with GitHub Actions, establishing baseline quality metrics measured by JaCoCo code coverage analysis

Data Engineering Intern, Vision Software Solutions – Charleston, SC

Sept 2022 – Jan 2023

- Containerized ETL pipeline for credit scoring algorithm using Docker and deployed infrastructure using Terraform Infrastructure as Code, improving system reliability and enabling consistent deployments
- Optimized MongoDB queries by reducing over-fetching and restructuring aggregation pipelines, improving credit scoring algorithm load times by 11% and reducing database query execution time

Education

College of Charleston – BS in Computer Science, GPA 3.8/4.0

May 2024

Certifications

AWS SAA-C03 – AWS Solutions Architect - Associate (expected Aug 2026)

Jul 2026